

Element E

While making the mirror, a member of our group had the idea of installing tin foil as a diffuser to make the LEDs appear spread out rather than coming from any certain source. We also decided on an infrared sensor for LEDs.

Tinfoil was used to diffuse the light made by the LEDs inside the mirror. This works because of the reflective properties of tinfoil. When light from the LEDs hits the tinfoil, it gets scattered in various directions due to the shape of the tinfoil. This effect causes the light to evenly distribute across the surface of the mirror, preventing the spots where the LEDs are to be obvious and make for a more spread illumination. This creates a softer and more even light within the mirror. Also we incorporated an infrared sensor. An infrared sensor works by detecting infrared radiation released by things. The sensor releases infrared light and when something comes into its path some of the emitted light is reflected back toward the sensor. The sensor sees the reflected infrared light and converts it into an electrical signal. The strength of this signal is relative to the amount of infrared radiation detected allowing the sensor to determine the presence or absence of objects within its detection range. In our mirror project, we utilized the infrared sensor to detect when someone uses the remote control, activating lights inside the mirror. We had professionals look at our project when a group of engineering students came in. They reviewed the prototype of the mirror and liked it. They commented on the weaker LED we used during the time and inquired how would the LEDs would be able to be controlled. They also questioned the shape and design of the mirror. The feedback of the prototype helped develop the mirror to what it is today, which includes brighter LEDs, an infrared sensor, and a more unique look using 3D-printing techniques.